

Phil R. Van-Lane

PhD Candidate, University of Toronto

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Profile

I am a PhD candidate (graduating Summer 2027) in the David A. Dunlap Department of Astronomy & Astrophysics at University of Toronto, and a Visiting Graduate Student at University of California San Diego. My research is focused on developing statistical, machine learning, and AI methodologies to study variability in low mass stars, with the goal of improving age constraints for exoplanet hosts. I have been actively engaged in service roles throughout my degree, having served as Co-President of the Graduate Astronomy Students Association in 2023-2024 and many other roles. Before beginning my PhD, I worked for seven years in software development, data engineering and product management.

Education

University of Toronto

Toronto, ON, Canada

PhD, Astronomy & Astrophysics

2022–2027

Thesis Advisors: Prof. Joshua Speagle, Prof. Gwendolyn Eadie, Prof. Ryan Cloutier (McMaster University)

Thesis Title: *From Stars to Statistics: Examining Exoplanet Populations through Stellar Activity with Machine Learning.*

Visiting Graduate Student, UC San Diego

San Diego, CA, USA

UCSD Host: Prof. Chris Theissen

2025–2027

University of Waterloo

Waterloo, ON, Canada

BSc, Earth Science (Geophysics specialization)

2010–2015

Thesis Advisor: Prof. Tony Endres

Thesis Title: *An Examination of GPR Processing and its Effect on Data Interpretation.*

Research

Refereed Publications

1. **Van-Lane, P. R.** et al., 2026, “EncoTESS: Age-Sensitive Encodings from Raw TESS Light Curves” (*submitted*), <https://arxiv.org/abs/2608.25019>.
2. Xia, A., **Van-Lane, P. R.** et al., 2026, “EmberFlow: Extending ChronoFlow for M dwarf Gyrochronology” (*in prep*).
***This is an undergraduate project for which I am the primary supervisor.
3. **Van-Lane, P. R.** et al., 2025, “ChronoFlow: A Data-Driven Model for Gyrochronology”, *ApJ*, 986, 59, <http://doi.org/10.3847/1538-4357/adcd73>.
4. **Van-Lane, P.**, Speagle, J. S., & Douglas, S. 2023, “A Novel Application of Conditional Normalizing Flows: Stellar Age Inference with Gyrochronology” (proceedings from the 2023 ICML Workshop on Machine Learning for Astrophysics), <https://arxiv.org/abs/2307.08753>.
5. Wagg, T., Broekgaarden, F., and **Van-Lane, P. R.**, (*in review*), “Streamlining and standardizing software citations with The Software Citation Station”.

6. Lu, Y., Pinsonneault, M., Ting, Y.S., **Van-Lane, P. R.** et al., 2026, “Anchoring Stellar Age Indicators: A Cross Calibration of [C/N] and Gyrochronology Ages via the Age-Velocity Dispersion Relation”, *AJ*, 171, 38, <http://doi.org/10.3847/1538-3881/ae196b>.
7. Tohuvavohu, A., et al. (25 co-authors incl. **Van-Lane, P.**), (*in review*), “LUVCam: A high-performance, low-cost, UV/optical camera for the future of astronomy in space”, <https://arxiv.org/abs/2607.28831>.
8. Khandelwal, A., et al. (9 co-authors incl. **Van-Lane, P.**), 2024, “Beyond CCDs: characterization of sCMOS detectors for optical astronomy” (proceedings from the 2024 SPIE Astronomical Telescopes + Instrumentation Workshop), <https://arxiv.org/abs/2409.16449>.

Students Supervised

- Alan Xia (Undergraduate, UC San Diego) 2026
- Prajakta Saraf (Undergraduate, UC San Diego) 2026

Presented Talks and Posters

1. Invited talk, [Université de Montréal Astrophysics Seminar](#) October 2026
2. Contributed talk, [Ensenada-San Diego Astrophysical Society Meeting](#) September 2026
3. Contributed talk, [Cool Stars 23: Open Clusters as Laboratories for Cool Star Evolution](#) June 2026
4. Contributed talk, [Empowering Science in the Data-Rich Era of Astronomy](#) April 2026
5. Contributed talk, [247th AAS Meeting](#) January 2026
6. Contributed poster, [Astronomical Data Analysis Software & Systems](#) November 2025
7. Contributed talk, [Joint Statistical Meetings](#) August 2025
8. Contributed talk, [9th TESS/16th Kepler Asteroseismic Science Consortium](#) July 2025
9. Contributed talk, [246th AAS Meeting](#) June 2025
10. Contributed talk, [Statistical Society of Canada Annual Meeting](#) May 2025
11. Invited talk, [UCSD “SMASH” Series](#) March 2025
12. Invited talk, [CfA Seminars](#) October 2024
13. Contributed talk, [8th TESS/15th Kepler Asteroseismic Science Consortium](#) July 2024
14. Contributed poster, [The 22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun](#) June 2024
15. Contributed poster, [CASCA AGM](#) June 2024
16. Contributed talk, [Globular Clusters and Their Tidal Tails](#) May 2024
17. Contributed talk, [Great Lakes Clusters and Streams](#) August 2023
18. Contributed talk, [7th TESS/14th Kepler Asteroseismic Science Consortium](#) July 2023
19. Contributed poster, [ICML Workshop on Machine Learning for Astrophysics](#) July 2023
20. Contributed talk, [CASCA AGM](#) June 2023

Scholarships and Funding

Doctoral Scholarships and Fellowships (\$184,500 total funding awarded)

NSERC Postgraduate Scholarship (PGS-D); \$120,000 over 3 years	2024–2027
Walter C. Sumner Memorial Fellowship; \$18,000 over 2 years	2025–2026
Lachlan Gilchrist Fellowship Fund; \$4,000	2024–2025
QEI Graduate Scholarship in Science and Technology; \$15,000	2023–2024
Robert Shelley Graduate Scholarship; \$5,000	2023–2024

NSERC Graduate Scholarship (CGS-M); \$17,500	2022–2023
Department of Astronomy & Astrophysics Entrance Award; \$5,000	2022–2023

Other Doctoral Awards, Funding, and Recognitions

Winner, Student Research Oral Presentation Award, SSC Annual Meeting	2025
Finalist, Astrostatistics Interest Group Student Paper Competition	2025
Recipient of Visiting Researcher Program funding, Center for Astrophysics	2024
3rd Place, CoolStars22 Early Career Researcher Poster Competition	2024
"You Lead, We Follow" Dunlap Institute Award	2024

Undergraduate Scholarships and Awards

Dean's Honours List x8 (every term)	2010–2015
Logan Student Prize	2015
Suncor Emerging Leaders Award in Science	2011–2015
J.P. Bickell Foundation Scholarship	2011–2015
Queen Elizabeth II Aiming for the Top Scholarship	2011–2015
Brantford Lapidary and Mineral Society Inc. Scholarship	2012–2013
UW President's Research Award	2012
NSERC Undergraduate Research Award	2012
Phillip Hall of Memorial Scholarship	2012
Canadian Exploration Geophysical Society Foundation Scholarship Award	2012
UW President's Scholarship of Distinction	2010–2011
Conestoga Rover's and Associates Entrance Scholarship	2010–2011

Service and Academic Leadership

Organizations, Conferences and Workshops

Web Director, Astrostatistics Interest Group (x2)	2026–2027
• Special Interest Group of the American Statistical Association. • Responsible for website redesign and maintenance.	
Co-chair (SOC and LOC), Software Citation Workshop	2026
• Organized four day workshop bringing experts together to define and implement best practices for software citation in astronomy.	
Organizing Committee, STATSTRO Workshop	2024–2025
• Helped to coordinate content and scheduling for both 2024 and 2025 versions of this workshop designed to bring astronomers and statisticians together. • Designed and developed the STATSTRO 2025 website.	

Department Leadership

**DADDAA refers to roles held with the David A. Dunlap Department of Astronomy & Astrophysics*
**GASA refers to roles held with the Graduate Astronomy Students' Association at U of T*

Co-President, GASA	2023–2024
• Negotiated a $\approx 10\%$ raised to the guaranteed funding package for graduate students, and an increased top-up for major awards. • Managed the activities of all GASA committees, coordinated prospective student visits, and aligned GASA feedback for a faculty search.	
Training and Mentoring Committee, DADDAA	2022–2023
• Organized and coordinated training and mentorship activities for all department members.	

Mediation Committee, GASA (x2)	2022–2025
• Developed and conducted a survey on financial health of GASA members. • Used the survey results to provide recommendations to the department for funding adjustments.	
Other GASA Positions	
Web Director	2025–2026
Graduate Student Mentor (x2)	2023–2025
Social Committee (x2)	2022–2025
Graduate Student Mentorship Coordinator	2022–2023

Teaching

Team Project Instructor, Data Sciences Institute Certificates	2024–2026
• This is a program designed to rapidly upskill individuals (with an emphasis on recent immigrants) who have employment experience but are looking to transition into a machine learning or data science career. • In my 2-week module (which I have delivered 5 times), I focus on developing the participants' "softer" skills required to excel in a technical career, such as self-organizing teamwork and delivering clear value to stakeholders.	
Head TA Positions (UofT)	
AST201: Stars and Galaxies, Head Tutorial TA	2024
• Coordinated duties for graduate and undergraduate TAs across the entire course. • Ensured feedback and evaluation was consistent from all TAs for projects and testing.	
AST101: The Sun and Its Neighbours, Head TA	2024
• Designed tutorial content and structure for ≈30 graduate TAs to deliver to ≈1500 undergraduate students over the duration of the course. • Ran weekly meetings with all of the TAs to make sure everyone was aligned on the course content and delivery.	
Other Graduate TA Positions (UofT)	
AST201: Stars and Galaxies (x3)	2023–2025
AST101: The Sun and Its Neighbours (x2)	2022–2023
Undergraduate TA Positions (UW)	
EARTH460: Applied Geophysics II	2015
EARTH260: Applied Geophysics I	2013
EARTH359: Flow Through Porous Media	2012

Community Engagement and Outreach

Co-organizer, AAS Workshop on Strategies for Rural Astronomy Education and Outreach	2027
• This is a single day workshop that I am co-organizing and will be co-instructing as part of the AAS 249 meeting in January 2027, titled "Small Towns, Big Skies: Strategies for Rural Astronomy Education and Outreach".	
Astronomy Educator, British Columbia Highschools	2024–2025
• Participated in two panels (for different schools) discussing career options in astronomy, and in STEM in general. • Presented an introductory astronomy classroom talk to Grade 6 students.	
Volunteer Coordinator, AstroTours	2022–2023
• Coordinate ≈20 volunteers for this monthly outreach event which includes a public talk and interactive astronomy activities.	
Volunteer, Astronomy On Tap Toronto	2022–2024
• Engaged members of the public and assisted with logistics for these events which are the largest astronomy outreach events in Toronto.	
Volunteer, AstroEDU Toronto	2023
• Helped facilitate this conference designed to address common issues in astronomy education and share experiences among astronomy educators.	

Previous Employment

Hootsuite Inc.

Product Manager, Data Technology

2021–2022

- Developed the product roadmap for the company's entire data engineering department.
- Managed an initiative to standardize all of the company's data pipelines.

Royal Bank of Canada

Senior Product Manager, Data Innovation Lab

2020–2021

- Led the technical implementation of the reporting tool for RBC's global "Return To Premises" (post-COVID) initiative.
- Developed the product roadmap for my team's in-house data visualization application.

Product Manager, Data Innovation Lab

2019–2020

- Expanded the reach of my team's internal data visualization product into several departments across the company.

Senior Developer, Data Innovation Lab

2018–2019

- Full stack developer.
- Built the first web application to be deployed on RBC's internal cloud.

Technical Systems Analyst, Corporate Real Estate IT

2016–2018

- Developed and automated ≈ 20 data pipelines for the Corporate Real Estate team.
- Deployed software to devices (eg. self-service kiosks, meeting room booking panels) throughout RBC office buildings.

Tetra Tech EBA

Geotechnician

2016

- Conducted materials testing and quality control for various construction projects at remote sites throughout Northwest Territories and Nunavut.
- Configured a remote alerting system for seismic vibrations induced during construction work at a hospital.

Co-op Jobs (while at University of Waterloo)

Geophysicist, Nexen Energy ULC

Jan–Aug 2014

Geotechnician, Tetra Tech EBA

May–Sep 2013

Researcher, UW Geophysics Lab

Sep–Dec 2012

Geologist, Lake Shore Gold Corp.

Jan–Apr 2012

Geologist, Lake Shore Gold Corp.

May–Aug 2011